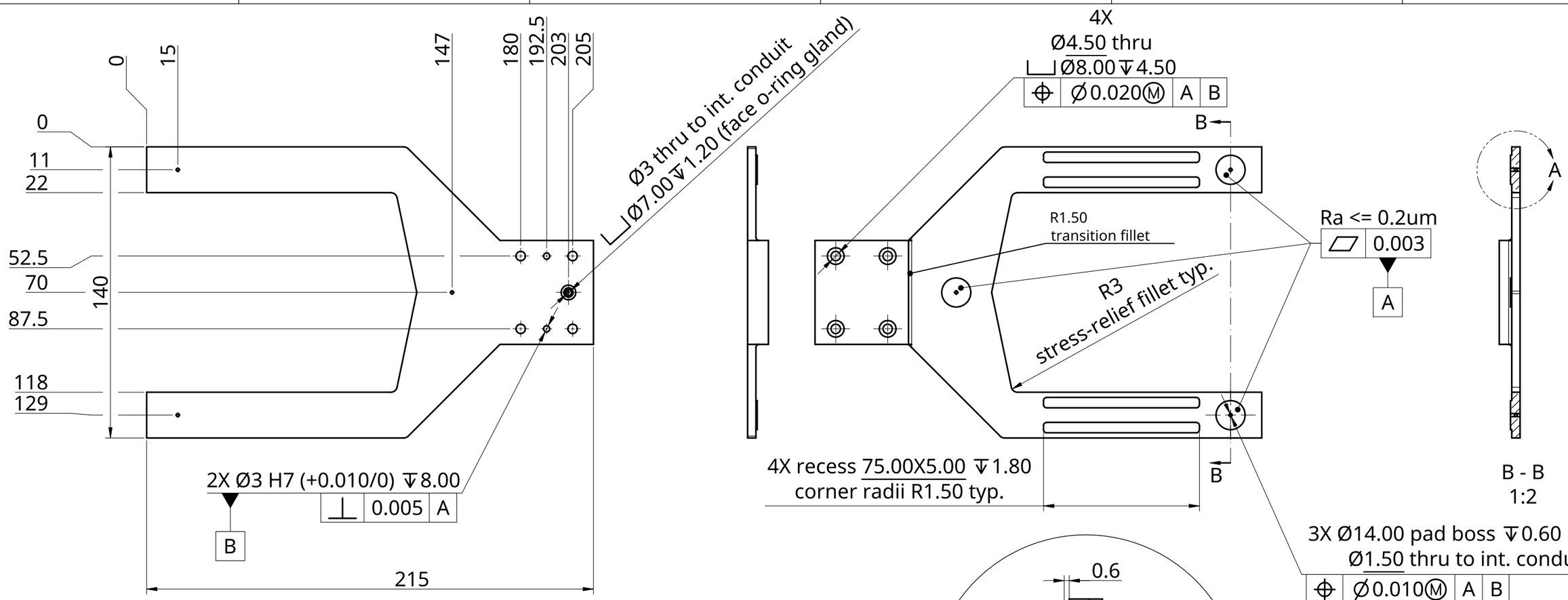


NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (WUA_END_EFFECTOR_RevA.STEP) GOVERNS NOMINAL GEOMETRY.
   1.3 GENERAL TOLERANCES: LINEAR +/- 0.05 mm, ANGULAR +/- 0.2 DEG (PER ISO 2768-f).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & THERMAL POST-PROCESSING:
   2.1 PRIMARY MATERIAL: ADDITIVE GRADE Ti-6Al-4V ELI (GRADE 23) PER ASTM F136.
   2.2 HOT ISOSTATIC PRESSING (HIP): 920°C +/- 10°C AT 120 MPa FOR 180 MINS UNDER ARGON.
   2.3 MICROSTRUCTURE: DENSE AND HOMOGENEOUS ALPHA+BETA NEEDLE MARTENSITE; DENSITY > 99.98%.
   2.4 CALCULATED NET MASS: 148.0 g (VOLUME: 33,408 mm^3).

3. SURFACE ENGINEERING & MACHINING:
   3.1 3X SUCTION PADS (DATUM A) COPLANARITY <= 0.003 mm; SURFACE FINISH Ra <= 0.2 MICRONS.
   3.2 NON-CRITICAL FLANK SURFACES: Ra <= 1.6 MICRONS VIA MAGNETIC ABRASIVE FINISHING.
   3.3 NO RESIDUAL POWDER, WELD SPLATTER, OR INTERNAL CONDUIT OCCLUSION PERMITTED.
   3.4 INTERNAL CONDUIT ISOLATION WALL THICKNESS MUST EXCEED 1.25 mm THROUGHOUT.

4. ULTRA-HIGH VACUUM & CLEANLINESS COMPLIANCE:
   4.1 COMPONENT COMPLIES WITH SEMI E49 / SEMI E57 ULTRA-HIGH VACUUM OUTGASSING LIMITS.
   4.2 NO HYDROCARBONS, ADHESIVES, ELASTOMERS, OR EMBEDDED SLAG RESIDUES PERMITTED.
   4.3 VACUUM LEAK RATE: GLOBAL HELIUM LEAK RATE < 1.0E-10 mbar*L/s PER ASTM E498.

5. CLEANROOM HANDLING & PACKAGING:
   5.1 ULTRASONIC DEGREASING IN REAGENT-GRADE IPA FOLLOWED BY HIGH-PURITY DI WATER RINSE.
   5.2 INSPECTED AND PACKAGED IN ISO CLASS 5 (FED-STD-209E CLASS 100) CLEANROOM ENVIRONMENT.
   5.3 DOUBLE-BAG SEALED UNDER PURGED HIGH-PURITY NITROGEN (N2) IN ANTISTATIC PE ENVELOPE.
   5.4 EACH PACKAGE MUST BE PERMANENTLY LABELED WITH DWG NO, LOT NO, AND CLEANROOM DATE.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS ANGULAR = ± ° SURFACE FINISH		NAME	SIGNATURE	DATE
	DRAWN			2026-10-12
	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
THIRD ANGLE PROJECTION	MATERIAL: Ti-6Al-4V ELI (Grade 23) per ASTM F136		FINISH	

TITLE: LITHOGRAPHY WAFER HANDLER HIGH-DYNAMICS MONOLITHIC END-EFFECTOR

SIZE	DWG NO.	REV.
A3	WUA_END_EFFECTOR_REVA	A

SCALE	WEIGHT	SHEET
NTS	0.148 kg	1 of 1